# Towards a General Theory of Medium-Sensitive Current-Mediated Ephaptic Coupling

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**Abstract**

In this note, the authors provide a general approach to three dimensional modeling of axon tracts in given extracellular media, with a view to capturing their extracellular-current mediated ephaptic interaction.

In [1] the authors remarked that extracellular medium anisotropy must affect ephaptic coupling. They noted that the inter-axonal medium is not isotropic. Thus the $W$-matrix [2] should be a matrix of vectors, not scalars. Suppose the function $f(\theta)$ represents the non-isotropy of the extracellular medium conductance. When $f(\theta)$ is large, then if there is an axon at that $\theta$, the link will be stronger. So each $W_{ij}$ must be proportional as well to the norm of the angular separation between $\theta_{max}$ and $\theta$ where $\theta_{max}$ is the argument that maximizes $f(\theta)$ and $\theta$ is the angle of the vector from axon $i$ to axon $j$.

In the present note, the authors make the further remark that the theory presented in [2] is a special case of a deeper theory that takes into account any axon configuration whatsoever. Here we will do well to recall that each axon in that theory was accompanied by one angle of inclination to the tract axis. This limits consideration to only those axons which lie in planes that themselves contain the $z$-axis.

If we want to consider the most general axon orientations, we must allow for two angles to represent each axon. One angle will be with the $xz$ plane and the second will be with the $yz$ plane. The partial derivative relation that was used to extend the cable equation depended on a single angle formulation. In future work we will need to extend it to accommodate the most general case described herein. This will have ramifications for synchronization, demyelination, tortuosity and many other derivative considerations. Once this deeper theory is ready, the medium-sensitivity discussed above will also need to be generalized and would perhaps necessitate a move from $W$ matrices to $W$ tensors.

## References

[1] Aman Chawla. *Ephaptic Information Flow.* Amazon KDP, 1 edition.

[2] Aman Chawla, Salvatore Morgera, and Arthur Snider. On axon interaction and its role in neurological networks. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 2019.